SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Engineer to Excel

COURSE OUTCOME COVERED

CO1: *Apply R Programming fundamentals including data structures, vectors, matrices, arrays and class types to perform mathematical operations and manage data effectively. (BL: 3)*

ASSESSMENT TOOL 1: Assignment 1

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks: 50** **Duration: 1 Hour**

Code of Conduct:

I S. Harish *(Name)* 192521003 *(Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

S. Harish

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding						
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer						
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application						
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response						
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand						
					Total per Question						

1.) Student mark Analyzer.

write a R program to analyze the marks of 10 students stored in a vector

* use for loop
* use if else
* total number in category.

Program

```
marks <- c(82,45,67,91,38,76,54,49)
distinction <- 0
pass <- 0
fail <- 0
for(m in marks){
  if(m >= 75){
    print(paste(m, "- Distinction"))
    distinction <- distinction + 1
  } else if(m >= 50){
    print(paste(m, "- pass"))
    pass <- pass + 1
  } else {
    print(paste(m, "- Fail"))
```

```
fail <- fail + 1 }}

cat ("Distinction", distinction, "\n")
cat ("pass:", pass, "\n")
cat ("Fail:", fail, "\n")
```

Output:

```
82 - Distinction
45 - Fail
67 - pass
91 - Distinction
38 - Fail
76 - Distinction
54 - pass
49 - Fail
88 - Distinction
72 - pass
Distinction : 4
pass: 3
Fail : 3
```

2.) ATM withdrawal challenge

* write an R program to stimulate an ATM with draw system

* balance with ₹1000
* 0 when user enters
* while() loop
* insufficient Balance
* invalid Amount.

Program

```
balance <- 10000
while (TRue) {
    amount <- as.numberic (readline
("Enter withdrawl amount (0 to stop:"))
if (amount == 0) {
  break
} else if (amount < 0){
  cat ("Invalid Amount\n")
} else if (amount > balance){
  cat ("Insufficient Balance\n") }
```

```
else if (amount %% 100 != 0){
    cat("Invalid Amount\n")
}else{
    balance <- balance - amount
    cat("Remaining Balance : Rs.", balance)
}
}
cat("Transaction ended.\n")
cat("Final Balance : Rs.", balance, "\n")
```

Output

```
Enter withdrawal amount : 5000
Remaining balance : 5000

Enter with drawal amout : 5000
Remaining balance : 0
Transaction ended
Final Balance : 0
```

3.) Matrix Diagonal Processing

* Given a square matrix, write an R program to process its elements?

* use nested for() loops
* main diagonal
* Secondary diagonal

Program:

```
m <- matrix(c(5, 2, 8,
              1, 7, 4,
              6, 3, 9), nrow = 3,
            byrow = TRUE)

n <- nrow(m)
main_sum <- 0
secondary_sum <- 0
cat("main diagonal element:")
for (i in 1:n) {
  for (j in 1:n) {
    if (i == j) {
      cat(m[i, j], " ")
```

```
main_sum <- main_sum + m[i,j]
}
if (i+j == n+1){
    secondary_sum <- secondary_sum
}}}
cat("\n main diagonal sum:", main_sum,
"\n")
cat("secondary diagonal sum:", secondary
-sum, "\n")
if (main_sum > secondary_sum){
    cat("main diagonal has the gerater.)
} else if (secondary.sum > main_sum){
    cat("secondary diagonal has the")
    else {
    cat("Both diagonal Sums are equal.\n")
}
```

Output

main diagonal element : 5 7 9
Secondry diagonal element : 9 7 6
main diagonal Sum : 21
Secondary diagonal Sum : 21
Both are equal

4. Looping over a List of Employee Records

Create an R list containing Info about 5 employees

* for() loop
* greater than 50,000
* each department.
* incomplete employee records.

Program

employee <- list(

list (name = "Arun", dept = "CSE"
, salary = 55000),

```
dept_count <- c()
highest_salary <- 0
highest_name <- ""

for (emp in employees) {
    if (is.null(emp$name) || is.null
        is.null(emp$salary)) {
        next
    }
    if (emp$salary > 5000) {
        cat("
        highest_salary <- emp$salary
        highest_name <- emp$salary
    }}
cat("In department-wise employee")
print(dept_count)
cat("Highest salary: Rs.", highest, "")
```

Output:

Arun - RS. 55000

chitra - RS. 62000

Deepa - RS. 51000

Department-wise employee: chitra

highest Salary : RS. 62000

5.) Create your own problem for coding challenge with solution.

```
attendance <-c (85, 60, 92, 70, 78)
for (a in attadence) {
    if (a<75) {
        cat (a, "- not eligible \n")
    } else {
        cat (a, "- Eligibility \n") }}

cat ("\n Enter additional attendance
values (-1 to Stop): \n")
```

```
cat "(In Enter values (-1 to Stop ): \n")
while (True){
    a <- as.numerical (readlines ("A:"):
    if (a == -1){
        break }
    if (is. naca) || a<0 || a >100){
        cat (" Invalid attadence \n")
        next }
    attadence <- c(attadence, a) }
m <- matrix (attadence, a)
cat ("(n Attadence matrix: \n))
print(m)
cat (" Highest Attadence, max(at), "%.n");
```

Output

80 - Eligible

65 - not eligible

Attadence : 88

Attendance : -1

Matrix : 85 60 92

70 78 88

highest addadence : 92%.